

Practical - 28

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Compute the Euclidean distance between two Breeze vectors. Use it for nearest neighbor classification.

Code:

```
build.sbt  Scala.scala x
1  import breeze.linalg._
2
3  ▶ object Scala extends App
4  {
5      val A = DenseVector(2.0, 3.0)
6      val B = DenseVector(5.0, 7.0)
7
8      val distance = norm(A - B)
9
10     println("Euclidean Distance: " + distance)
11
12     val C = DenseVector(2.5, 3.5)
13
14     val d1 = norm(C - A)
15     val d2 = norm(C - B)
16
17     if (d1 < d2)
18         println("Nearest Neighbor: A")
19     else
20         println("Nearest Neighbor: B")
21 }
```

Output.

```
Euclidean Distance: 5.0
Nearest Neighbor: A

Process finished with exit code 0
```

